

PRALAYADARSHI

Compound Risk → Lead Time → Action

Hyperlocal Flash Flood & Debris Flow Early Warning with Ridge Evacuation Intelligence

GREEN: Nominal

YELLOW: Watch

ORANGE: Prepare

RED: Evacuate

ORGANIZATION & DOMAIN CONTEXT

Organization: Ministry of Home Affairs | NDRF / Disaster Management Division

Problem Statement: PS 26192 — Flash Flood Prediction System for Hilly Regions

Theme / Category: Disaster Management • Software Category

Pilot Basin Focus: Chamoli & Rudraprayag Districts, Uttarakhand (Himalayan Catchments)

THE NORTH STAR LEAD TIME

35 Mins

Estimated Lead Time Window

- Transforms chaotic panic into structured, orderly village evacuation.
- Dynamic physics calculation based on river surge rate of rise & rainfall spikes.
- Directly triggers plain-language Citizen Card dispatch to local feature phones.
- Routes families to high-ground shelters strictly outside gorge flood plains.

Proposed Solution & North Star Architecture

Moving from Fragmented District Advisories to Hyperlocal, Actionable Decision Support

CURRENT SYSTEM SHORTCOMINGS (THE PROBLEM)

- ✗ Macro-level alerts covering 100+ km² with zero village precision.
- ✗ Isolated data silos: Rainfall, river gauges, and slope geology never fused.
- ✗ Zero evacuation guidance: Citizens told 'heavy rain expected' without safe paths or shelter assignments.
- ✗ Widespread alert fatigue from chronic district-wide false alarms.

THE PRALAY INNOVATION (OUR SOLUTION)

- ✓ Hyperlocal Catchments: Village-level watershed monitoring (Raini, Tharali, Joshimath).
- ✓ Deterministic Physics Fusion: River rate-of-rise + rainfall intensity + soil pore saturation.
- ✓ Dynamic Lead-Time Engine: Explicit 'Estimated Lead Time' in minutes (e.g. 35 mins).
- ✓ Active Evacuation Routing: Automated shelter assignment avoiding gorge flood corridors.

1. WHERE IS THE RISK?

Hyperlocal village catchment polygons (Lat/Lon, basin terrain, slope angle).

2. HOW SEVERE?

Compound Risk (0–100) mapped to 4-Tier Scale: GREEN / YELLOW / ORANGE / RED.

3. HOW SOON?

Explicit 'Estimated Lead Time' in minutes (e.g. 35 mins), projected from surge rise.

4. WHAT NOW?

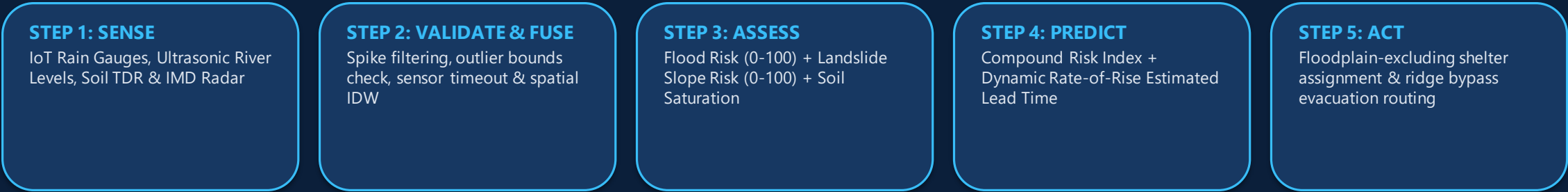
Designated safe shelter above flood line + turn-by-turn ridge bypass trail.

DUAL-INTERFACE PARADIGM • Authority Operations Dashboard: Live GIS multi-hazard overlays, sensor telemetry, aggregate district metrics, and manual broadcast overrides.

CITIZEN ALERT CARD (PRD §18) • Citizen Emergency Alert Card (PRD §18): Jargon-free vernacular SMS/Card — 'EVACUATE IMMEDIATELY to Raini Primary School via Upper Ridge Trail. Avoid River Ghat.'

Technical Approach & Multi-Source Fusion Pipeline

Physics-Grounded Deterministic Computation + Multi-Source Sensor Ingestion Pipeline



DETERMINISTIC COMPOUND RISK FORMULATION (PRD §9 & §10)

Compound Risk = $\max(\text{Flood}, \text{Landslide}) \times 0.65 + \min(\text{Flood}, \text{Landslide}) \times 0.20 + \text{Vulnerability} \times 15$

Prioritizes the dominant hazard surge while compounding secondary slope failure risks and village population exposure.

- Flood Factor (0-100): River level + rate of rise + rainfall accumulation
- Landslide Factor (0-100): Saturation pore pressure + slope steepness + trigger rain

ENGINEERING SAFETY & ACCURACY GUARDRAILS

- 🛡️ **Hardware Validation:** Sensor timeout marks source UNAVAILABLE, never silently assumed 0.
- 🛡️ **Alert Fatigue Suppression:** 30-minute cooldown suppression for recurring alerts; immediate auto-escalation on severity spike.
- 🛡️ **Hydraulic Routing:** Shelters located inside active flood plains are strictly excluded from assignment.

PRODUCTION ARCHITECTURE & CORE TECHNOLOGY STACK

Python FastAPI | PostgreSQL + PostGIS | TimescaleDB Hypertables | Redis & Celery | ESP32 / MQTT Telemetry | Pydantic V2 | React 18 + Vite | Tailwind + TypeScript

Async Non-blocking API Core • PostGIS Spatial Geometries • TimescaleDB Telemetry Hypertables • Edge IoT Telemetry

Feasibility, Viability & Implementation Roadmap

Engineering Rigor, Edge-Resilient Deployment & Phased Operational Rollout

Phase A: Pilot Prototype

Months 1–3 (Delivered & Verified)

Chamoli & Rudraprayag pilot catchments, 14/14 automated tests passing, 12-step cloudburst simulator, React decision panes.

Phase B: Field Telemetry & IoT

Months 4–7

Solar-powered ESP32 nodes, ultrasonic stream sensors, LoRaWAN gateways across Alaknanda & Mandakini basins.

Phase C: State & Regional Scale

Months 8–12

Integration with SDMA Uttarakhand, CWC hydrological feeds, multi-lingual vernacular SMS dispatch.

FIELD RISK ASSESSMENT & RESILIENCE MITIGATION MATRIX

- ▲ **IMD API Outage / Network Cut:** Local station caching + Inverse Distance Weighting (IDW) spatial fallback to adjacent active nodes.
- ▲ **Field Sensor Physical Failure:** Heartbeat watcher flags source UNAVAILABLE; model degrades gracefully to upstream rain gauge estimates.
- ▲ **Mountain Cellular Dropouts:** LoRaWAN edge mesh buffering up to 72 hours of readings on SD flash; emergency alert relays.
- ▲ **False Alarm Alert Fatigue:** Dual-threshold confirmation + 30-minute zone cooldown suppression; instant escalation on worsening.

ECONOMIC VIABILITY

₹15,000

Avg. CapEx per Solar IoT Monitoring Node

- 10x more affordable than imported seismic/Doppler systems.
- Solar + LiFePO4 battery with 5-day cloudy backup.
- Uses open standards (MQTT / LoRaWAN / GSM 4G).

Impact, Operational Benefits & Target Metrics

Quantifiable Life-Safety Outcomes for Mountain Communities & Disaster Responders

30–45 Mins

Estimated Lead Time

Target window for pre-impact evacuation

< 10%

False Positive Rate

Target threshold to eliminate warning fatigue

< 5%

Missed Event Target

Zero-compromise disaster detection target

> 0.90

F1-Score Target

Balanced precision-recall performance goal

NDRF & SDMA COMMANDERS

Common Operational Picture: Exact village catchments at risk, automated shelter capacity tracking, and pre-staged rescue deployment.

DISTRICT MAGISTRATES & OFFICIALS

Defensible Decision Support: Scientific lead-time estimates and audit-ready automated incident logs for accountability.

VULNERABLE CITIZENS & GRAM PANCHAYATS

Clear Plain-Language Instructions: Plain SMS alerts naming safe high-ground shelters and avoiding lethal river corridors.

DETECT EARLIER. IDENTIFY LOCALLY. ACT IN TIME.

Research, References & Technical Grounding

Grounded in Authoritative Hydrological Standards, Geotechnical Formulations & National Datasets

1. DATASETS & TELEMETRY FEEDS

- India Meteorological Department (IMD) AWS & Doppler Radar Rainfall
- Central Water Commission (CWC) River Gauge Baselines
- NASA SRTM 30m Digital Elevation Models (Slope, Aspect, Drainage)
- Survey of India & Census 2011 Village Spatial Boundaries

2. GEOTECHNICAL & HYDROLOGICAL MODELS

- Deterministic Infinite Slope Stability Formulation (Factor of Safety)
- SCS-CN Hydrological Runoff & Stream Power Index (SPI)
- Topographic Wetness Index (TWI) for Himalayan Debris Basins
- Ensemble Time-Series Sequence Modeling (LSTM / XGBoost)

3. DISASTER INVENTORIES & VALIDATION

- Geological Survey of India (GSI) National Landslide Susceptibility
- Uttarakhand SDMA 2013 & 2021 Post-Disaster Forensic Archives
- Chamoli Rishi Ganga Flash Flood Historical Reconstruction Data
- EM-DAT International Disaster Database Records

4. PROTOCOLS & GOVERNANCE ALIGNMENT

- National Disaster Management Authority (NDMA) Standard Operating Protocols
- WMO Early Warnings for All (EW4All) Global Guidelines
- Sendai Framework for Disaster Risk Reduction (2015–2030) Target G
- TRAI Guidelines for Priority Emergency Telecom Warning SMS